

The effects of price competition on firms' operations and market price: Evidence from a retail gasoline market

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ARTICLE INFO

JEL codes:

L52

L81

Q41

Keywords:

Price competition

Entry effect

Thrifty gas station policy

Operational probability

Gasoline price

Retail gasoline market

ABSTRACT

We study the effects of price competition induced by entries of thrifty gas stations on incumbent gas stations' operational probabilities and gasoline prices. Using administrative panel data of the entire population of retail gas stations in South Korea, we document that entries of thrifty gas stations decreased incumbent gas stations' operational probability but marginally reduced gasoline prices. Our heterogeneity analysis indicates that the operational probability impacts were greater when the market competition was already intense or when the pre-entry profit margins were low. The previous studies mainly focused only on price impacts of market competition. This study provides novel evidence that price competition might not always decrease market prices if incumbent firms stop operating their business. The results imply that a more complete understanding of the effects of price competition can be possible by jointly estimating its impacts on incumbent firms' operational probabilities and product prices.

1. Introduction

A competitive market price can achieve efficient resource allocations and improve social welfare. Several studies have investigated the relationship between competition and product prices in the retail gasoline market. In addition to market prices, incumbent firms can adjust other measures such as revenue in response to competitive price pressure.¹ Investigating firms' responses in other metric along with product price can enable us to have a more comprehensive understanding of the impacts of price competition.

The probability of a business operation can be a useful metric for studying the impact of price competition, especially in the long run. Price competition might not always significantly lower market prices because of heterogeneous underlying market conditions. If the market is not competitive, intensified competition will lower market prices. However, if the market is already competitive or the profit margins of incumbents are low, intensified competition may not significantly lower market prices. Instead, incumbents might have an incentive to stop operating their business or exit the market. This implies that the effect of price competition can be underestimated when only considering the

price impact and not the effects on business operations. However, few studies demonstrate the relationships of price competition incumbent firms' operations and product prices.

The purpose of study is to investigate the effects of price competition on incumbent firms' operational probability and gasoline prices in retail gasoline markets in South Korea (Korea). There were speculations that the oligopolistic vertical integrations of oil refiners and retail gas stations reduced market competition and charged higher retail gasoline prices than competitive prices. To increase market competition, the Korean government introduced a thrifty gas station policy in December 2011 and encouraged entries of thrifty gas stations with financial support. Since the thrifty gas stations were supplied oil products at cheaper prices by the government than their competitors, the entries of thrifty gas stations into retail gasoline markets were expected to induce price competition among incumbent retail gas stations.

For identification, we exploit spatial and temporal variations in the entries of thrifty gas stations induced by the policy. Since retail gasoline markets are local, we compare changes in operational probabilities and gasoline prices between incumbent gas stations experienced the entries of thrifty gas stations in the immediate neighborhoods (within a 1 km

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¹ Several studies examined the effects of market competition on other outcomes of interests such as employment (Ellickson and Grieco, 2013), product quality (Matsa, 2011; Forder and Allan, 2014), product assortment (Bauner and Wang, 2019), and revenue (Arcidiacono et al., 2020).

radius) to those experienced the entries of thrifty gas stations in surrounding neighborhoods (within a 1–2 km band) following [Hastings \(2004\)](#) and [Arcidiacono et al. \(2020\)](#). To implement this research design, we employ event study design and difference-in-differences (DID) models. Using administrative panel data on the entire retail gas stations' weekly gasoline prices and operational status over almost 10 years, we investigate whether the entries of thrifty gas stations in immediate neighborhoods decreased incumbent gas stations' operational probability and gasoline prices.

The effects of thrifty gas stations' entries can be heterogeneous based on the degree of market competition. In response to price competition, incumbent firms are more likely to stop operating a business if the market is already competitive, and thus, they could not further adjust gasoline prices. However, those incumbent gas stations are more likely to lower gasoline prices if the market is less competitive and thus they could further adjust gasoline prices. To test this conjecture, we presume that the intensity of price competition can increase if an additional thrifty gas station enters the market. Then we estimate the marginal effects of entries of thrifty gas stations.

The effects of price competition can be heterogeneous by the amount of profit margins. In response to price competition, incumbent gas stations with low profit margins find it difficult to reduce gasoline prices and have a stronger incentive to stop operating business. However, those with higher profit margins can decrease gasoline prices instead of lowering operational probability. Unfortunately, our data do not provide information on individual gas stations' profit margins. To alleviate this issue, we exploit the theoretical fact that incumbent firms in areas with high entry costs can enjoy higher profit margins than those in areas with low entry costs. We estimate heterogeneous effects by the locations because retail gas stations in metropolitan areas are required to pay higher rents (i.e. greater entry costs) than those in non-metropolitan areas.

In this study, we find that the entries of thrifty gas stations in immediate neighborhoods decreased incumbent gas stations' operational probability but marginally reduced gasoline prices. On average, the entry of a thrifty gas station decreased incumbent firms' probability of operating business by 1.6 percentage points and weekly gasoline price per liter by 2.4 KRW (0.8 US cent per gallon).² The effects of thrifty gas stations' entries are heterogeneous: In response to price competition, incumbent firms are more likely to stop operating a business if the market was already competitive, and thus, they could not further adjust gasoline prices. However, those incumbent gas stations are more likely to lower gasoline prices if the market was less competitive and thus they could further adjust gasoline prices.

The rest of this study is structured as follows: [Section 2](#) discusses the related literature. [Section 3](#) provides the background of Korea's gasoline industry and the policy about thrifty gas stations. [Section 4](#) describes the data. In [Section 5](#), we present our empirical strategy. [Section 6](#) shows the results, and [Section 7](#) concludes.

2. Related literature

There are numerous empirical studies on the gasoline retail market with different areas of focus including price dynamics (pass-through, asymmetric pricing, and Edgeworth cycles), mergers and regulations, and price dispersion ([Eckert, 2013](#)). Our study examines the relationship between local competition pressure and retail prices.³ Several studies

have measured competitive effects using variables such as the number of stations ([Van Meerbeeck, 2003](#); [Sen, 2003](#); [Ning and Haining, 2003](#); [Barron et al., 2004](#); [Eckert and West, 2005](#); [Hosken et al., 2008](#); [Kihm et al., 2016](#); [Bergantino et al., 2020](#)), station density measured by the number of stations per square kilometer or per 1000 inhabitants ([Clemenzen and Gugler, 2006](#); [Pennerstorfer, 2009](#)), and concentration measured according to the concentration ratio and Herfindahl-Hirschman Index ([Sen, 2003](#); [Eckert and West, 2005](#); [Hosken et al., 2008](#); [Kihm et al., 2016](#); [Clemenzen and Gugler, 2006](#)) in a station's radius or in an administrative district. The findings are mixed, but most studies report that the number of gas stations or station density is negatively associated with price. However, [Sen \(2003\)](#) and [Hosken et al. \(2008\)](#) found that the relationship is not statistically significant, and [Pennerstorfer \(2009\)](#) noted a positive relationship.

Other studies examined the effects of unbranded or independent stations on price. [Van Meerbeeck \(2003\)](#) examined the effect of independent station presence on price, and [Clemenzen and Gugler \(2006\)](#) and [Pennerstorfer \(2009\)](#) investigated the effect of the local market share of independent (unbranded) stations on price. The former two studies found that the presence and proportion of independent gas stations had a negative relationship with price, but the latter documented insignificant associations. [Hastings \(2004\)](#), [Hastings \(2010\)](#), and [Taylor et al. \(2010\)](#) examined the effects of conversions from thrifty to regular gas stations on the gasoline prices of nearby gas stations in the Los Angeles and San Diego metropolitan areas. The authors found mixed evidence of the gasoline price impacts of conversions to a regular gas station using different data sources.

This study contributes to the literature in two folds. First, we provide new evidence on the effects of price competition in retail gas station markets. The previous studies mainly focused on the price impacts of market competition and reported mixed evidence. Although [Eckert and West \(2006\)](#) documented that a greater presence of independent gas stations in the local market induces the shutdown of major brand gas stations, there are few attempts to use the results to reconcile the mixed evidence of price impacts. This study provides new insights by investigating how incumbent firms jointly decide their operations and prices in response to market competition: price competition can lower incumbent gas stations' operational probability without significant price adjustments. In addition, this study also investigates how incumbent gas stations' responses are heterogeneous by underlying market conditions and their pre-entry profit margin levels. We argue that this analysis enables us to have a more complete understanding of why price competition does not always decrease gasoline prices in retail gasoline markets.

Second, this study provides evidence on the long run effects of price competition in retail gas station markets. Previous studies mainly focused on the short run price impacts of market competition. We aim to complement these studies by using almost 10 years of panel data of individual gas stations, and to provide new evidence for the dynamic effects of price competition. Since incumbent gas stations' operation and price adjustment can differently evolve over time after entries of thrifty gas stations, the results of this study can shed new light on the effects of market competition.

3. Background

3.1. Retail gasoline market in Korea

The oil refining industry in Korea has formed an oligopoly comprising four oil refiners, SK Energy, GS Caltex, S-OIL, and Hyundai

² As of August 9, 2021, 1 KRW is about 0.0009 USD (0.09 cent). Since 1 gallon is 3.78541 liter, we convert gasoline price per liter into US cent per gallon by multiplying 0.09 and 3.78541 to our estimates.

³ We do not have information on the wholesale oil price charged by the Korea National Oil Corporation to thrifty gas stations. Thus, in this study, we cannot conduct a pass-through analysis like [Doyle and Samphantharak \(2008\)](#) and [Marion and Muehlegger \(2008\)](#).

Oilbank. According to Petronet, the market share of the 4 brands in 2008 measured by the number of gas stations was 97.1% of the total gasoline stations (i.e., 36.8% for SK, 27.8% for GS Caltex, 18.5% for Hyundai Oilbank, and 14% for S-OIL), while the share of independent gas stations was only 2.9%. In 2018, the market share of the 4 brands decreased to 87.8% and the share of independent gas stations increased to 12.2%. The increase in the number of independent gas stations during this period was mainly due to the increase in the number of thrifty gas stations since 2011, which accounted for 80% of independent gas stations in 2018.

These four refiners exerted market power in distribution and retail markets via vertical integration (KPMG Korea, 2015). Besides, they also had exclusive contracts with privately-owned gas stations together with a post-settlement system.⁴ The oligopolistic market structure, along with vertical integrations with retail gas stations, could have discouraged competition in the retail gasoline market and resulted in higher gasoline prices than competitive prices.

Oil refiners could gain dominant bargaining power in brand contracts with private gas stations because of the lack of competitiveness induced by independent gas stations (Jung, 2012). In the pre-treatment period, the average gasoline prices of independent gas stations and branded gas stations were 1782.7 KRW per liter and 1806.1 KRW per liter, respectively (December 2009–November 2011). Despite the price competitiveness of independent gas stations, as of 2010, the proportion of independent gas stations in Korea was 4.5%, far lower than that of other countries.⁵ This can be attributed to the fact that independent gas stations were not a suitable alternative to brand gas stations for consumers at that time because of illegal behaviors such as the use of low-quality gasoline and deception in the amount of oil. In addition, it was difficult for independent gas stations to obtain a stable supply of oil products because they did not have a brand contract with an oil refiner (Kim and Kim, 2011).

3.2. Thrifty gas station policy

In December 2011, the Korean government implemented the thrifty gas station policy to induce price competition in the retail gasoline market. The Korean government sponsored independent gas stations via the following channels. First, to resolve consumer distrust, the government introduced the public brand of independent gas stations called “thrifty gas stations,” which provide oil products with the same quality as that of other brands' gas stations. Second, to ensure a stable supply, the Korea National Oil Corporation (KNOC) supplies thrifty gas stations with oil products, and its supply price is lower than that of other refiners. Specifically, KNOC purchases a large amount of oil cheaply in an auction from the four oil refiners and supplies it to thrifty gas stations without a margin.⁶ Third, to force thrifty gas stations to sell oil at lower prices, KNOC terminates their contracts if they do not charge sufficiently lower prices than their nearby competitors. As a result, thrifty gas stations charged lower gasoline prices by 29–37 KRW per liter (9.88–12.61 US cent per gallon) compared to their competitors from 2013 to 2019 (Korea National Oil Corporation, 2020).

Fig. A1 shows that the number of thrifty gas stations has increased

⁴ The post-settlement system refers to a method in which the gas station orders the oil refiner without knowing the exact price of the oil and is notified of the final price from the oil refiner two weeks later or at the time of monthly settlement. Since it was hard for retail gas stations to select oil refiners with cheaper prices, this practice could minimize competition among refiners. The Korea Fair Trade Commission took corrective action against these actions in 2009.

⁵ Choi and Hwang (2012) provided the proportion of independent gas stations in other countries as of 2009: Japan (24.8%); and as of 2010, France (45.8%), Germany (69.1%), the UK (44.7%), and the US (48.6%).

⁶ Along with KNOC, the Nonghyup Agricultural Cooperative Federation Agribusiness Group and Korea Expressway Corporation participate as partners in the thrifty gas station policy.

since it was first introduced in December 2011. The number of thrifty gas stations increased rapidly for the first three years after its introduction and gradually increased afterward. Their locations are geographically dispersed in metropolitan areas. In 2019, the number was over 1100, reaching 10% of the total number of gas stations in Korea. The share of petroleum products sold by thrifty gas stations 16.9% as of November 2019 (Kim, 2020).

4. Data

Since 2008, the KNOC established the Oil Price Information Network (OPINET) for providing information on the daily gasoline price of all retail gas stations in Korea and for tracking that information up to date. We created a binary indicator of individual gas stations' operational status based on the price information: It takes 1 if information on the gasoline price of a gas station is provided, and 0 otherwise. In addition, the OPINET provides additional information on gas stations' characteristics such as type (self-service vs staff-service) and brands (SK Energy, GS Caltex, S-OIL, Hyundai Oilbank, and thrifty gas stations). For the empirical analysis, we construct individual gas station level weekly panel data on the average gasoline prices and operational status during December 2009 and June 2019.⁷ We excluded gas stations located on highways because their relevant geographic markets are likely larger than those of gas stations located in cities from the sample. Table A1 provides summary statistics of dependent variables used in this study. In an average week, 80% of the total incumbent gas stations engage in business operations. Their average gasoline price is around 1694.1 KRW per liter (5.77 USD per gallon). Furthermore, about 28% of incumbent firms operate self-service gas stations in one week.

To construct treatment variables, we used the information on the exact location of all gas stations (i.e., address) provided by the OPINET. First, we calculated linear distances between an incumbent gas station and all thrifty gas stations.⁸ Second, for each incumbent gas station, we count the number of thrifty gas stations within a distance band. Table A2 shows the average number of thrifty gas stations by distances. On average, an incumbent gas station had experienced 0.33, 0.58, 0.43, 0.49, and 0.55 entries of thrifty gas stations within a 1 km, 1–2 km, 2–3 km, 3–4 km, and 4–5 km band, respectively.

5. Empirical strategy

To study the effects of entries of thrifty gas stations on incumbent gas stations' likelihood of operating business and gasoline price, we exploit temporal and spatial variations in entries of thrifty gas stations. Since the timings of entries of thrifty gas stations are different across incumbent gas stations, we normalize the entry timing to zero when the first thrifty gas station enters the local market for each incumbent gas station. Then, we investigate how incumbent gas stations' operational

⁷ Since the data covers the population of retail gasoline stations in South Korea, it required significant computing time for estimations when using the daily level data. To avoid this practical issue, we used data at the weekly level.

⁸ The linear distance measure, which is used to define a local market, is relatively easier to construct because it does not require extensive computation or extra data. To the authors' knowledge, there are two alternative measures, namely driving distance and driving-time distance, that can be used to define a local market. First, we calculate linear distance instead of driving distance because of computing time. However, existing studies demonstrate that linear distances can be a reasonable proxy for driving distance (Boscoe et al., 2012). Second, Houde (2012) suggests driving-time distance based on actual commuting path by newly incorporating the fact that consumers choose a retail gasoline station based on their commuting paths. We acknowledge that the driving-time distance based on commuting paths is the most realistic measure by which to define a local market. However, it is difficult to access the commuting path data; thus, we could not apply that type of method to define a local market.

probabilities and gasoline prices evolve before and after the entry of the first thrifty gas station within a 1 km radius (treatment group). As the entries of thrifty gas stations are geographically dispersed, other factors affecting the operation of incumbent gas stations and gasoline prices could be spatially distributed as well. Assuming that the distributions of these factors are smooth over geographical space, we chose incumbent gas stations that experienced entries of thrifty gas stations within a nearby neighborhood—i.e., a 1–2 km band—as a control group following Hastings (2004). To implement this research design idea, we consider the following event study design specification, while restricting the sample period between 100 weeks before entry to 200 weeks after the entry:

$$y_{i,w,t} = \sum_{w=-100}^{200} \beta_w \times \text{FirstEntry}_w \times \text{Entry}^{1km}_i + \mu_i + \lambda_t + \gamma_{m(i)}(t) + \varepsilon_{i,w,t}, \quad (1)$$

where i and t indicate an individual incumbent gas station and week; $y_{i,w,t}$ is the outcome variables of interest; Entry^{1km}_i is 1 if an incumbent gas station i 's nearest thrifty gas station is located within a 1 km radius, and 0 if the incumbent gas station i 's nearest thrifty gas station is located within a 1–2 km band; FirstEntry_w represents binary indicator of a week w since the entry of the first thrifty gas station; λ_t and μ_i represent time fixed effects and incumbent fixed effects, respectively⁹; $\gamma_{m(i)}(t)$ indicates an incumbent gas station i 's market-specific linear trends, which is approximated by district-specific linear trends. β_w s are parameters of interest that capture the dynamic effects of entries of thrifty gas stations within a 1 km radius compared to those of entries of thrifty gas stations within a 1–2 km band. β_w presents the estimated difference in outcome variables between the treatment and control groups in week w ($w > 0$) compared to the same difference in the week of the first thrifty gas station's entry ($w=0$). This implies that we estimate the effects of the entry of the first thrifty gas station after w weeks. If the entry of thrifty gas stations increased competition in the local retail gasoline market, then incumbent gas stations could stop operating their business or reduce gasoline prices after the entry. This means β_w s ($w > 0$) could be negative if the entry of thrifty gas stations induced significant market competition. For statistical inference, we calculated standard errors clustered at the gas station level.¹⁰

For the identification of β_w s, the entries of thrifty gas stations are required to be exogenous to changes in outcomes of interests, conditionally on μ_i , λ_t , and $\gamma_{m(i)}(t)$ (Arcidiacono et al., 2020). However, thrifty gas stations may enter a market after investigating changes in market conditions. This could cause a bias in our estimates. If a thrifty gas station entered a market where incumbent gas stations had already been decreasing (or increasing) operational probabilities or gasoline prices, it could overemphasize (or underemphasize) the negative effects of entries of thrifty gas stations on those outcome variables. As an indirect test for the identification assumption, we examine if trends of operational probabilities and gasoline prices are parallel before the entry of the thrifty gas stations following Arcidiacono et al. (2020). We closely examine if the estimated β_w s (for $w < 0$) are small in magnitude and statistically insignificant.

Then, we exploited temporal and spatial variations in the number of entries of thrifty gas station. In particular, we estimate following a DID model.

$$y_{i,t} = \sum_{j=1}^J \beta_j \text{NumEntry}^j_{i,t} + \lambda_t + \mu_i + \gamma_{m(i)}(t) + u_{i,t}, \quad (2)$$

where we follow the same notations as in Eq. (1). $\text{NumEntry}^j_{i,t}$ represents the number of thrifty gas stations that newly entered within a $j-1$ to j km band of an incumbent gas station i at week t . In the baseline analysis, we used two distance band groups such as 0–1 km and 1–2 km. Then, β_1 and β_2 captured the effects of one unit increase in thrifty gas stations within a 1 km radius (for $j=1$) and a 1–2 km band (for $j=2$). If the entry of thrifty gas stations increased market competition, then incumbent gas stations could stop operating their business or reduce gasoline prices after that entry. This implies that β_1 and β_2 are negative. Since competition is significant when a thrifty gas station enters an immediate neighborhood, the magnitude of β_1 is likely greater than that of β_2 . Although we consider a 2 km radius to define local markets, the results are robust when estimating the effects of entries of thrifty gas stations in a shorter distance such as a 1 km radius and in broader distances such as a 3–5 km radius. In this study, we report the results of using a 5 km radius as an alternative definition of the local market to save space. To account for serial correlation in the dependent variables within an incumbent gas station, we calculate standard errors clustered at the gas station level.

Then we conduct heterogeneity analysis to examine potential mechanisms. First, the effects of entries of thrifty gas stations can be associated with the intensity of price competition. Under the presumption that the degree of price competition increases when the number of thrifty gas stations in immediate neighborhoods increases, we consider the following regression model:

$$y_{i,t} = \sum_{j=1}^2 \beta_j^{1st} \text{Entry}^j_{i,t} + \sum_{j=1}^2 \beta_j^{2nd} \text{Entry}^j_{i,t} + \lambda_t + \mu_i + \gamma_{m(i)}(t) + u_{i,t}, \quad (3)$$

where we split $\text{NumEntry}^j_{i,t}$ of Eq. (2) into $1^{st} \text{Entry}^j_{i,t}$ and $2^{nd} \text{Entry}^j_{i,t}$ ($j=1, 2$). Each regressor indicates the entries of the first and second thrifty gas stations, respectively, within a $j-1$ to j km band of an incumbent gas station i .¹¹ Otherwise, we follow the same notations as in Eq. (1). β_j^1 and β_j^2 are parameters of interests which represent the marginal effects of entries of thrifty gas stations. Since market competition intensifies as more thrifty gas stations enter within a $j-1$ to j km band, the magnitude of β_j^2 can be greater than that of β_j^1 given j . Because market competition can be severe in a nearby neighborhood, the magnitudes of the effects can be greater when $j=1$. To account for serial correlation in the dependent variables within an incumbent gas station, we calculate standard errors clustered at the gas station level.

Next, the effects of entries of thrifty gas stations can be heterogeneous based on pre-entry profit margins. In response to price competition, incumbent gas stations with higher profit margins are likely to reduce gasoline prices instead of stopping operating business. However, those with lower profit margins might decrease the operational probability instead of bearing profit losses. Unfortunately, our data do not provide information on individual gas stations' profit margins. To alleviate this limitation, we exploit the fact that firms in areas with high entry costs are likely to exploit higher profit margins. Retail gas stations in metropolitan areas generally pay higher entry costs due to higher rents, and thus, they could enjoy higher profit margins than those in non-metropolitan areas.¹² Based on this presumption, we estimate differences in the effects of entries of thrifty gas stations between

⁹ Since retail gas stations (other than thrifty gas stations) can enter or exit the market, the composition of incumbent gas stations can change over time. To control for a sample selection bias due to changes in sample composition, we included in the baseline analysis gas station fixed effects. As an additional check, we also restricted the sample to those who operated over the entire sample period, finding that the results remained intact.

¹⁰ Statistical inferences remain robust when calculating alternative standard errors clustered at the district level, an alternative definition of the local market.

¹¹ The maximum number of thrifty gas stations within a 1 km radius is 3. However, its share is only 0.6% of the entire sample, and thus, it is hard to precisely estimate the marginal effects of the third entry of thrifty gas station. Thus, we do not create separate dummy variables indicating the third entry of the thrifty gas station when modifying the regression model.

¹² As proxies for relative market supply, the numbers of gas stations per 100,000 KRW (90 US dollar) per capita income, per 10,000 persons, and 10,000 cars between metropolitan areas and other areas are 9.4 and 38.5, 1.2 and 3.6, and 3.7 and 9.3, respectively. This shows that the relative numbers of gas stations are greater in non-metropolitan areas than in metropolitan areas.

metropolitan and non-metropolitan areas by employing a triple-differences (TD) specification:

$$y_{i,t} = \sum_{j=1}^J \beta_j \text{NumEntry}_{i,t}^j + \sum_{j=1}^J \beta_j^M \text{Metropolitan} \times \text{NumEntry}_{i,t}^j + \lambda_t + \mu_i + \gamma_{m(i)}(t) + u_{i,t}, \quad (4)$$

where we add $\text{Metropolitan} \times \text{NumEntry}_{i,t}^j$ ($j=1, 2$) to regression Eq. (2), and Metropolitan is a dummy variable indicating if an incumbent gas station locates in Seoul, Busan, Daegu, Incheon, Gwangju, Daejeon, and Ulsan. Otherwise, we follow the same notations as in Eq. (1). Then, β_j ($j=1, 2$) represents that the effects of an entry of a thrifty gas station in non-metropolitan areas, while β_j^M ($j=1, 2$) captures the additional effects of an entry of a thrifty gas station in metropolitan areas compared to those in non-metropolitan areas. If the entries of thrifty gas stations are less (more) likely to reduce operation probability (gasoline prices) in an area with higher entry costs, then β_j^M could be positive (negative). Because market competition can be severe in a nearby neighborhood, the magnitudes of the effects can be greater when $j=1$. To account for serial correlation in the dependent variables within an incumbent gas station, we calculate standard errors clustered at the gas station level.

6. Empirical results

6.1. Effects of entries of thrifty gas stations

First, we study how incumbent gas stations' operational probabilities and gasoline prices evolved before and after the entry of thrifty gas stations, using the event study model in Eq. (1). Fig. 1 shows the event study estimates (darker points) of operational probability and gasoline prices, along with their 95% confidence intervals (indicated in caps) in Panels A and B, respectively. Panel A presents that the estimated operational probability impacts are close to zero and statistically insignificant before the entry of a thrifty gas station, which provides evidence supporting the parallel trend assumption during the pre-entry period. The entries of thrifty gas stations did not immediately lower incumbent gas stations' likelihood of operating business. The operational probability decreased between 40 and 50 weeks after the entry of a thrifty gas station. The estimates show little evidence that the reduced operational probability would rebound afterward.¹³ We then examined whether the reduced likelihood of operations led to an exit by modifying the operation indicator to equal one if we never observe a price from an incumbent gas station after period t . Fig. A2 demonstrates that the entries of thrifty gas stations did not significantly increase incumbent gas stations' exit probability. According to the Ministry of Trade, Industry and Energy of Korea, only half the retail gas stations that temporarily stopped operations exited the market between 2015 and 2020. Combined with this fact, our results imply that the majority of incumbent gas stations temporarily stop their operations in response to increased competition in the local market.¹⁴

¹³ Unfortunately, to the authors' knowledge, there is no convincing economic theory or empirical evidence on when incumbent gas stations stop operations in response to increased local competition. Instead, we ensured there were no other events that coincidentally occurred in the two years post-treatment. It is difficult to formally test joint relationships between operational probability and gasoline prices in our event study design framework. Thus, we speculate their relationships based on the observed patterns in Fig. 1 (Panels A and B).

¹⁴ In this study, we could not investigate the behavior of incumbent gas stations during the temporal business closure because of data limitations. We leave this as an avenue for future research.

Panel B shows the dynamic effects of the entry of the thrifty gas station on gasoline prices. The estimates before the entry are positive and constant. This implies that incumbent gas stations kept charging higher gasoline prices before the thrifty gas stations enter the market.¹⁵

Then they decreased gasoline prices immediately before the entries of thrifty gas stations. Findings from Panels A and B indicate that incumbent gas stations' dynamic responses to the entries of thrifty gas stations could be heterogeneous over time. In the short run, the entries of gas stations decreased gasoline prices with null impacts on the operational probability, but in the long run, the entries of thrifty gas stations lowered the operational probability without further reductions in gasoline prices.

As more gas stations stop operating business, incumbent gas stations that still operate business have an incentive to revert gasoline prices in the long run. To test this conjecture, we re-estimated the effects of the entries of thrifty gas stations on gasoline prices by restricting the sample to those who never stop operations during the sample period. Appendix Fig. A3 shows that price impacts are like those of panel B of Fig. 1: Incumbent gas stations sharply decreased gasoline prices immediately before the entries of thrifty gas stations. They reverted gasoline prices after 50 weeks of a thrifty gas station's entry, which coincided with the timing when incumbent gas stations decreased the operational probability. However, they did not fully revert gasoline price to pre-entry level, probably because of the raging price competition induced by thrifty gas stations.

Then, we exploit spatial and temporal variations in the number of entries of thrifty gas stations using Eq. (2). Panel A of Table 1 shows DID estimates of the average effect of an entry of a thrifty gas station on the operational probability. Column (1) shows that an entry of a thrifty gas station within a 1 km radius decreased the probability of operating business by 1.6 percentage points, which is statistically significant at the 1% level. However, an entry of a thrifty gas station within a 1-2 km band did not reduce incumbent gas stations' likelihood of operating business. The estimates are small in magnitude and statistically insignificant. Column (2) indicates that the results remain robust when adding district-specific quadratic trends as a more flexible control for market-specific trends. Column (3) presents that entries of thrifty gas stations within a 2-5 km band did not reduce gas stations' likelihood of operating business.¹⁶ Since 80% of the entire incumbent gas stations operate in an average week (Table A1), the 1.6 percentage points reduction in operational probability is an approximately 2% reduction from the average operation probability, of 80%.

Panel B shows DID estimates of the effects of an entry of a thrifty gas station on weekly gasoline price. Column (1) shows that an entry of a thrifty gas station within a 1 km radius and a 1-2 km band decreased weekly gasoline price per liter by 2.4 KRW (0.8 US cent per gallon) and 1.1 KRW (0.37 US cent per gallon), respectively. The estimates are statistically significant at the 1% and 5% levels, respectively. Column (2) indicates that the results are robust when adding district-specific quadratic trends. Column (3) that there was no additional price impact of the entries of thrifty gas stations longer distances. Since the

¹⁵ Since we examined the effects of price competition induced by entries of thrifty gas stations in immediate neighborhoods, we argue that incumbent gas stations could expect the timing of those entries.

¹⁶ Table A3 indicates that the entries of thrifty gas stations did not increase the probability of incumbent gas stations' exit. The estimates are small in magnitude and statistically insignificant.

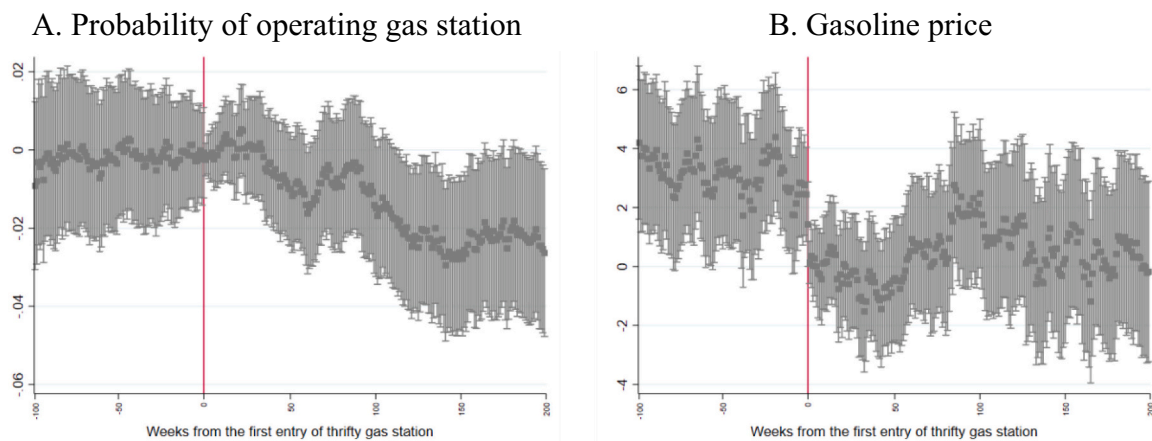


Fig. 1. Dynamic effects of entry of a thrifty gas station.

Data source: OPINET, December 2009 – June 2019.

Notes: Gray squares represent the estimated effects of the introduction of thrifty gas stations on the probability of operating business. We include gas station fixed effects, year-month-week fixed effects, and district-specific linear trend as controls. Standard errors are clustered at the gas station level. Caps indicate 95% confidence intervals.

Table 1
The effects of entries of thrifty gas stations.

	(1)	(2)	(3)
A. Dependent variable: Pr(operating business)			
# Thrifty stations w/in a 1 km radius	−0.016*** (0.006)	−0.017*** (0.006)	−0.016*** (0.006)
# Thrifty stations w/in a 1–2 km band	0.006 (0.004)	0.005 (0.004)	0.006 (0.004)
# Thrifty stations w/in a 2–3 km band			−0.004 (0.004)
# Thrifty stations w/in a 3–4 km band			−0.001 (0.004)
# Thrifty stations w/in a 4–5 km band			0.000 (0.004)
Observations	2,816,011	2,816,011	2,816,011
R-squared	0.639	0.641	0.639
B. Dependent variable: Gasoline price			
# Thrifty stations w/in a 1 km radius	−2.436*** (0.613)	−2.292*** (0.610)	−2.446*** (0.612)
# Thrifty stations w/in a 1–2 km band	−1.084** (0.460)	−1.041** (0.456)	−1.090** (0.458)
# Thrifty stations w/in a 2–3 km band			−0.076 (0.526)
# Thrifty stations w/in a 3–4 km band			−0.558 (0.498)
# Thrifty stations w/in a 4–5 km band			−0.769* (0.464)
Observations	2,256,300	2,256,300	2,256,300
R-squared	0.980	0.981	0.980
C. Dependent variable: Pr(operating self-service)			
# Thrifty stations w/in a 1 km radius	0.004 (0.004)	0.003 (0.004)	0.005 (0.004)
# Thrifty stations w/in a 1–2 km band	−0.002 (0.003)	−0.003 (0.003)	−0.002 (0.003)
# Thrifty stations w/in a 2–3 km band			0.009*** (0.003)
# Thrifty stations w/in a 3–4 km band			−0.001 (0.003)
# Thrifty stations w/in a 4–5 km band			0.008** (0.003)
Observations	2,256,300	2,256,300	2,256,300
R-squared	0.906	0.907	0.906
Radius used to define local market	2 km	2 km	5 km
District-specific linear trend	Y	Y	Y
District-specific quadratic trend		Y	

Data source: OPINET, December 2009 – June 2019.

Notes: We include gas station fixed effects and year-month-week fixed effects as additional controls. Standard errors are clustered at the gas station level. The notation *** indicates significance at 1% level, ** at 5% level, * at 10% level.

Table 2
Marginal effects of the entries of thrifty gas stations.

Dependent variables:	Pr(Operating Business)	Gasoline Price	Pr(Operating Self-serve)
	(1)	(2)	(3)
1st entry w/in a 1 km radius	−0.014** (0.006)	−3.062*** (0.653)	−0.002 (0.004)
2nd entry w/in a 1 km radius	−0.034* (0.019)	−0.008 (2.096)	0.039** (0.018)
1st entry w/in a 1-2 km band	0.005 (0.005)	−1.416*** (0.549)	−0.001 (0.003)
2nd entry w/in a 1-2 km band	0.012 (0.009)	−0.994 (1.158)	−0.002 (0.007)
Observations	2,816,011	2,256,300	2,256,300
R-squared	0.639	0.980	0.906

Data source: OPINET, December 2009 – June 2019.

Notes: We include gas station fixed effects, year-month-week fixed effects, and district-specific linear trends as controls. Standard errors are clustered at the gas station level. The notation *** indicates significance at 1% level, ** at 5% level, * at 10% level.

annual gasoline consumption per car of Korean households is about 1192 l as of 2018 (Korea Petroleum Association), the saved gasoline expenditure of an average household is about 2861 KRW (2.57 US dollar).

In response to increases in price competition, incumbent gas stations can adjust other measures to maintain profit margins as large as possible. We investigated if the entries of thrifty gas stations encouraged incumbent gas stations to convert to self-service gas stations. Since it is required to pay additional costs of hiring staff to operate staff-service gas stations, incumbent gas stations might switch to a self-service gas station to save the labor costs. DID estimates in Panel C shows little evidence that entries of thrifty gas stations increased the likelihood of operating self-service gas stations. Column (1) shows that the entry of a thrifty gas station within a 1 km radius increased the probability of operating a self-service gas station by 0.4 percentage points, which is statistically insignificant. The entry of a thrifty gas station within a 1–2 km band marginally reduced the likelihood of incumbent gas stations operating a self-service gas station by 0.2 percentage points, but the estimate is statistically insignificant. Column (2) shows that the results are robust when adding district-specific quadratic trends as a more flexible control for market-specific trends. Column (3) shows that entries of thrifty gas stations within a 2-5 km band did not significantly change the likelihood of operating a self-service gas station.¹⁷

6.2. Heterogeneity analysis

The baseline DID analysis provides consistent evidence that price competition induced by the entries of thrifty gas stations decreased incumbent gas stations' operational probabilities and marginally reduced gasoline prices. This provides novel evidence that price competition in the retail gas station market might not always decrease gasoline prices, like the findings of the previous studies that documented null effects of conversions of the thrifty gas station on gasoline prices in the U.S. (Hastings, 2010; Taylor et al., 2010). In this subsection, we study possible mechanisms using heterogeneity analysis.¹⁸

¹⁷ Although some estimates in column (3) are statistically significant, we do not argue that the estimates capture the effects of price competitions due to a lack of a systematic pattern across estimates.

¹⁸ We acknowledge that the mechanisms will not be exclusive and exhaustive.

Table 3
The effects of the entries of thrifty gas station entries by location.

Dependent variables:	Pr(Operating Business)	Gasoline Price	Pr(Operating Self-serve)
	(1)	(2)	(3)
# Thrifty stations w/in a 1 km radius	−0.022*** (0.006)	−1.637** (0.659)	0.000 (0.005)
# Thrifty stations w/in a 1-2 km band	0.006 (0.004)	−0.725 (0.494)	−0.002 (0.003)
Metropolitan × #Thrifty stations w/in a 1 km radius	0.023* (0.012)	−3.102** (1.528)	0.015 (0.010)
Metropolitan × #Thrifty stations w/in a 1-2 km band	0.001 (0.008)	−1.306 (1.096)	0.001 (0.007)
Observations	2,816,011	2,256,300	2,256,300
R-squared	0.639	0.980	0.906

Data source: OPINET, December 2009 – June 2019.

Notes: We include gas station fixed effects, year-month-week fixed effects, and district-specific linear trends as controls. Standard errors are clustered at the gas station level. The notation *** indicates significance at 1% level, ** at 5% level, * at 10% level.

6.2.1. Marginal effects of the entries of thrifty gas stations

The effects of entries of thrifty gas stations can be associated with the intensity of price competition. As Fig. 1 hinted, entries of thrifty gas stations decreased gasoline prices in the short run and the probability of operating business in the long run. One plausible explanation is that incumbent gas stations might have reduced gasoline prices when the price competition was relatively mild and stopped operating a business when the price competition was intensified. To further test this conjecture, we estimate the marginal effects of each entry of the thrifty gas station by using Eq. (3).

Table 2 summarizes estimated coefficient values of $1^{st}Entry_{it}^j$ and $2^{nd}Entry_{it}^j$ that capture the marginal effects of the entries of the first and second thrifty gas stations. Column (1) shows that the entry of the first thrifty gas station within a 1 km radius reduced incumbent gas stations' operational probabilities by 1.4 percentage points, and the entry of the second additionally decreased the probability by 3.4 percentage points. The estimates are statistically significant at the 5% and 10% levels, respectively. However, there was little effect of entries of thrifty gas stations within a 1-2 km band on the operational probability. Column (2) shows that the entry of the first thrifty gas station within a 1 km radius reduced incumbent gas stations' gasoline price by 3.1 KRW per liter (1.1 US cent per gallon), which is statistically significant at the 1% level. However, the entry of the second thrifty gas station within a 1 km radius did not additionally reduce gasoline price. It presents that the entry of the first thrifty gas station within a 1-2 km band decreased gasoline price by 1.4 KRW per liter (0.48 US cent per gallon), which is statistically significant at the 1% level. However, there were no additional reductions in gasoline prices after the entry of the second thrifty gas station. The results indicate that incumbent gas stations were likely to lower gasoline prices when price competition was relatively mild. However, they reduced the operational probability when price competition was intensified due to the additional entry of a thrifty gas station in an immediate neighborhood. We conjecture that they could not further decrease gasoline prices probably because their profit margins were already low. Column (3) shows that the entry of the first thrifty gas stations within a 1 km radius did not affect incumbent gas stations' likelihood of operating self-service gas stations. However, the entry of the second thrifty gas station increased the probability of operating self-service gas station by 3.9 percentage points, which is statistically significant at the 5% level. Nevertheless, there was little impact of the entries of thrifty gas stations within a 1-2 km band. The results imply that incumbent gas stations switched to self-service stations when the

price competition was intensified as additional thrifty gas stations entered immediate neighborhoods.

6.2.2. Effects of the entries of thrifty gas stations by location

The effects of entries of thrifty gas stations can be heterogeneous based on pre-entry profit margins. In response to price competition, incumbent gas stations with higher profit margins are likely to reduce gasoline prices instead of stopping operating business. However, those with lower profit margins might decrease the operational probability instead of bearing profit losses.¹⁹ To examine such conjecture, we estimate differences in the effects of entries of thrifty gas stations between metropolitan and non-metropolitan areas by employing a triple-differences (TD) specification in Eq. (4).

Table 3 provides evidence that is consistent with our conjectures. Column (1) shows that an entry of a thrifty gas station within a 1 km radius decreased the operational probability by 2.2 percentage points in non-metropolitan areas, which is statistically significant at the 1% level. The TD estimate shows that an entry of a thrifty gas station had positive effects on the operational probability in the metropolitan area. The estimate is 2.3 percentage points and is statistically significant at the 10% level. This implies that an entry of a thrifty gas station did not reduce the operational probability in metropolitan areas. We find little evidence that an entry of a thrifty gas station within a 1-2 km band decreased the operational probability regardless of locations. Column (2) presents that an entry of a thrifty gas station within a 1 km radius reduced gasoline prices by 1.6 KRW per liter (0.55 US cent per gallon) in non-metropolitan areas, and the estimate is statistically significant at the 5% level. The TD estimate presents that an entry of a thrifty gas station additionally reduced gasoline price by 3.1 KRW per liter (1.06 US cent per gallon) in metropolitan areas, and the estimate is statistically significant at the 5% level. There was little heterogeneous price impact of entries of thrifty gas stations within a 1-2 km band. Column (3) shows little evidence of the heterogeneous effects of an entry of a thrifty gas station on the probability of operating self-service gas station by location. The entry of a thrifty gas station within a 1 km radius and a 1-2 km band marginally changed the probability of operating a self-service gas station by 0.01 and 0.2 percentage points, and the estimate is statistically insignificant. The TD estimates indicate that the entry of a thrifty gas station additionally increased the probability of operating a self-service gas station by 1.5 and 0.1 percentage points within a 1 km and a 1-2 km bands in metropolitan areas, respectively. However, the estimates are statistically insignificant.

7. Conclusion

In this study, we examined the effects of price competition induced by the entries of thrifty gas stations on incumbent gas stations' operational probability and weekly gasoline prices. Using the administrative panel data of all retail gas stations in Korea, we reported new evidence that provides a more complete understanding of the effects of price competition and enables drawing useful policy implications.

First, we document that the entries of thrifty gas stations reduced incumbent gas stations' operational probabilities while marginally lowering gasoline prices. Several previous studies on the effects of retail

gasoline market competition provide mixed evidence of the price impacts of market competition (Eckert, 2013). To our knowledge, there have been few attempts to reconcile these findings. Since these studies have paid little attention to the operations behavior of incumbent gas stations, our study can provide new insights that market competition might not always lower gasoline prices if incumbent firms stop operating their business. We argue that our results can broaden understanding of the potential consequences of market competition induced by the entry of thrifty gas stations in local retail gasoline markets.

Second, our heterogeneity analysis indicates that the effects of market competition induced by entries of thrifty gas stations were heterogeneous in terms of the degree of market competitiveness and incumbent firms' pre-entry profit margins. The increase in access to information on local market conditions means that governments could obtain some measures about market conditions or market competitiveness before introducing a policy. Combined with this technological progress, our results imply that a more micro-managed policy design could enable a government to effectively achieve an intended policy goal while minimizing unintended consequences.

We acknowledge the limitations of this study. First, we could not provide explanations why gas stations would not exit the market, but instead, temporarily close the business. One possible explanation is that gas stations should pay additional costs when exiting the market instead of temporarily closing the business.²⁰ Second, in addition to the conversion to a self-service station, incumbent gas stations can adjust other factors such as the quality of service in response to price competition. Lastly, it might seem to be puzzling that operation impacts of entries of thrifty gas stations are relatively greater than price impacts. To better understand whether incumbent gas stations have an incentive to stop operating business in response to price competition, we should have been able to access to information on operating profits and gasoline sales quantity. Unfortunately, we could not examine the effects of the entries of thrifty gas stations on such metrics because of data limitations. We leave these as an avenue for future research.

CRedit authorship contribution statement

Kanghyok Koh: Conceptualization, Methodology, Software. **Sungho Jeon:** Conceptualization, Data curation, Resources, Visualization. **Jinhyuk Lee:** Conceptualization, Methodology, Supervision.

Acknowledgements

Koh acknowledges financial supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2021S1A5A2A03064205). Lee acknowledges financial support by the BK21 FOUR (Fostering Outstanding Universities for Research) funded by the Ministry of Education (MOE) of Korea and National Research Foundation (NRF) of Korea. The authors are grateful to the editor and two anonymous referees for their valuable suggestions and comments. The authors also thank seminar participants at Korea University, Yonsei University, KER International Conference, and monthly workshop of Korean Labor Economic Association for useful comments. All errors are our own.

Appendix A

¹⁹ Unfortunately, we cannot provide direct evidence that those in metropolitan areas have experienced higher pre-entry markups. Given this, we acknowledge that our heterogeneity results can be also explained by factors such as market size other than market competition. This limitation leads us to interpret the results with caution.

²⁰ For example, it is known that it costs about 140 million KRW (126,000 USD) to shut down a gas station: about 70 million KRW (63,000 USD) to dismantle the gas station, and about 70 million KRW (63,000 USD) to clean up the soil in case of soil contamination. According to the news, in large cities, gas station sites tend to be easily sold and used for other purposes, allowing them to easily be converted into market exit after temporary closures. However, in the case of gas stations in other regions, it is not easy to sell the site, so there is no way to raise the cost of exiting the market, so there are cases where they are left temporarily closed.

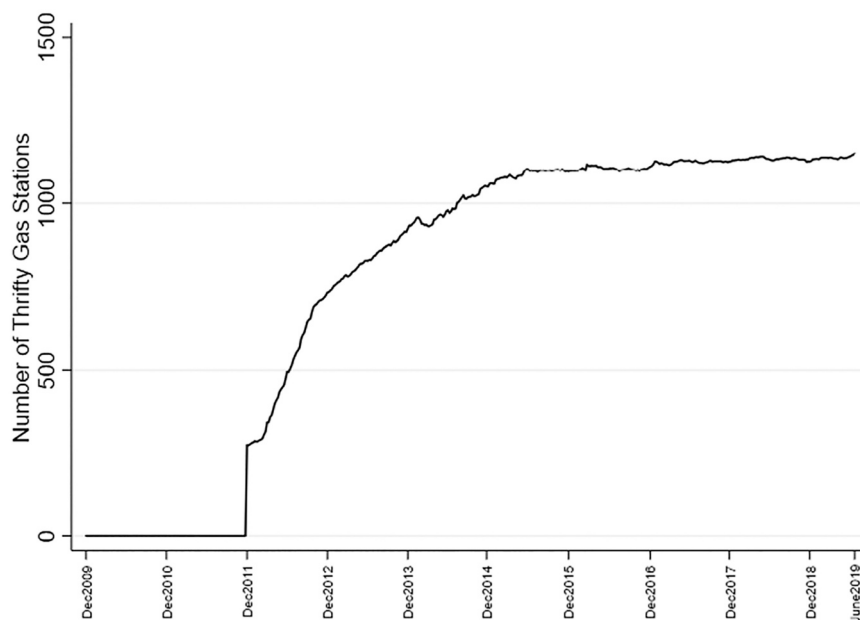


Fig. A1. Trends of the number of thrifty gas stations.
Data source: OPINET, December 2009 – June 2019.

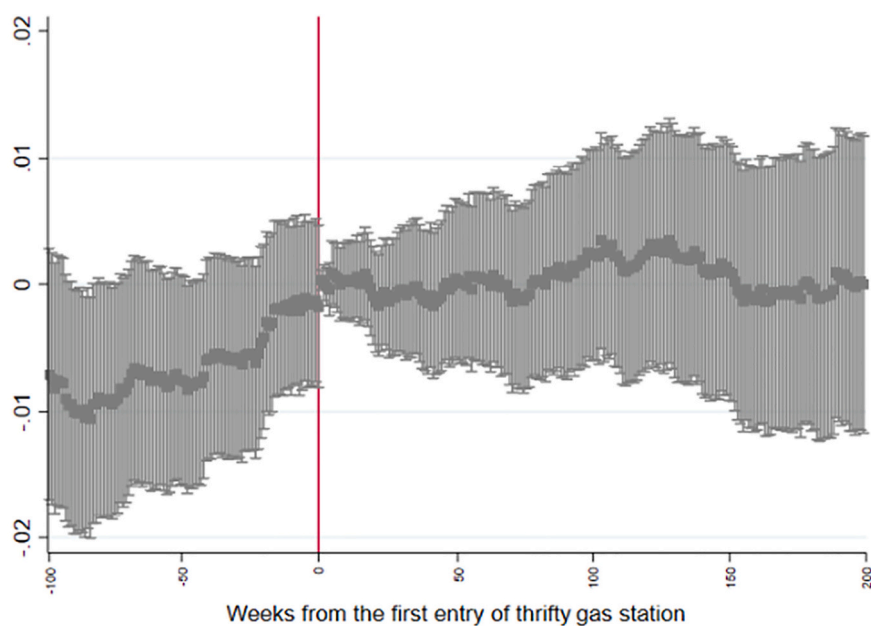


Fig. A2. Dynamic effects of entry of a thrifty gas station on the probability of incumbent gas stations' exit.

Data source: OPINET, December 2009 – June 2019.

Notes: Gray squares represent the estimated effects of the introduction of thrifty gas stations on the probability of exit. We include gas station fixed effects, year-month-week fixed effects, and district-specific linear trend as controls. Standard errors are clustered at the gas station level. Caps indicate 95% confidence intervals.

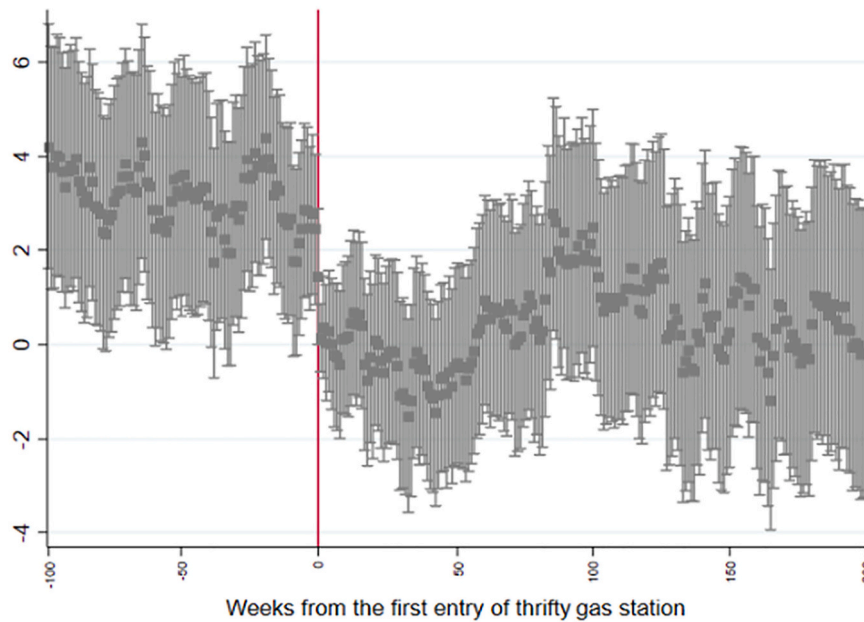


Fig. A3. Dynamic effects of entry of a thrifty gas station on gasoline price using incumbents that operated the entire sample period.

Data source: OPINET, December 2009 – June 2019.

Notes: Gray squares represent the estimated effects of the introduction of thrifty gas stations on the probability of operating business. We include gas station fixed effects, year-month-week fixed effects, and district-specific linear trend as controls. Standard errors are clustered at the gas station level. Caps indicate 95% confidence intervals.

Table A1

Summary of dependent variables.

	Mean	Standard Deviation
Pr(operating business)	0.80	0.40
Weekly gasoline price (KRW)	1694.1	221.2
Pr(operating self-service station)	0.28	0.45

Data source: OPINET, Dec 2011- June 2019.

Table A2

Summary of incumbent gas station's exposure to thrifty gas stations.

	Mean	Standard Deviation
#Thrifty $\leq$ 1 km	0.33	0.53
1 km < #Thrifty $\leq$ 2km	0.58	0.71
2 km < #Thrifty $\leq$ 3km	0.43	0.72
3 km < #Thrifty $\leq$ 4km	0.49	0.76
4 km < #Thrifty $\leq$ 5km	0.55	0.82

Data source: OPINET, Dec 2011- June 2019.

Table A3

The effects of entries of thrifty gas stations on the probability of exit.

	(1)	(2)	(3)
# Thrifty stations w/in a 1 km radius	0.004 (0.003)	0.005 (0.003)	0.004 (0.003)
# Thrifty stations w/in a 1-2 km band	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)
# Thrifty stations w/in a 2-3 km band			0.000 (0.002)
# Thrifty stations w/in a 3-4 km band			0.004 (0.002)
# Thrifty stations w/in a 4-5 km band			-0.000 (0.002)
Observations	2,816,011	2,816,011	2,816,011
R-squared	0.665	0.668	0.665
	0.004	0.005	0.004
District-specific linear trend	(0.003)	(0.003)	(0.003)
District-specific quadratic trend	-0.001	-0.001	-0.001

Data source: OPINET, December 2009 – June 2019.

Notes: We include gas station fixed effects and year-month-week fixed effects as additional controls. Standard errors are clustered at the gas station level. The notation *** indicates significance at 1% level, ** at 5% level, * at 10% level.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2022.105889>.

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